

TX4 Robustness Statistics Audit

Corrected blocker-antichain definitions, exact QMC/MC recomputation, and Vancouver handoff

Date: 2026-09-20 | Branch: research/tx4-robustness-statistics-audit

Executive verdict

The retained ExtraTrees tier failed the preregistered strict gate: 18 false-minimal conditions, 14 missed-minimal conditions, 0.925 exact-H4 agreement, and 0.702 H0-antichain agreement. The prespecified exact fallback was therefore executed for all 4,096 QMC and 5,000 MC coordinates across all 16 portfolios, totaling 145,536 exact DAE portfolio evaluations.

The legacy H4_PRESENT flag tested H4 instability rather than inclusion-minimal H4 membership. This complement/minimality bug produced 156 legacy-positive but corrected-negative conditions in the 426-condition exact calibration.

Final machine gate: PASS for all 32 invariants.

Definitions and provenance

For each condition, U0 is the set of portfolios with $\alpha_{EM} \geq 0$ in the frozen 0.3-1.5 Hz band. H0 is the inclusion-minimal antichain of U0. H4_UNSTABLE means H4 belongs to U0; H4_PRESENT means H4 belongs to H0; EXACT_H4 means H0 is exactly the singleton {H4}; NONCOMPOSABLE means H0 contains a blocker of size at least two; KAPPA is the smallest blocker size or NULL.

QMC results are fixed-design coverage fractions. MC results are assumed independent-uniform bounded-box probabilities with Wilson intervals. No U005/H005 threshold was found in the retained preregistration or code, so those endpoints were not invented or analyzed.

Corrected primary endpoints

Endpoint	QMC coverage	MC probability	MC Wilson 95%
H4_UNSTABLE	0.577881	0.581400	[0.567668, 0.595007]
H4_PRESENT	0.113770	0.115400	[0.106838, 0.124553]
EXACT_H4	0.113770	0.115400	[0.106838, 0.124553]
NONCOMPOSABLE	0.427246	0.436600	[0.422907, 0.450391]

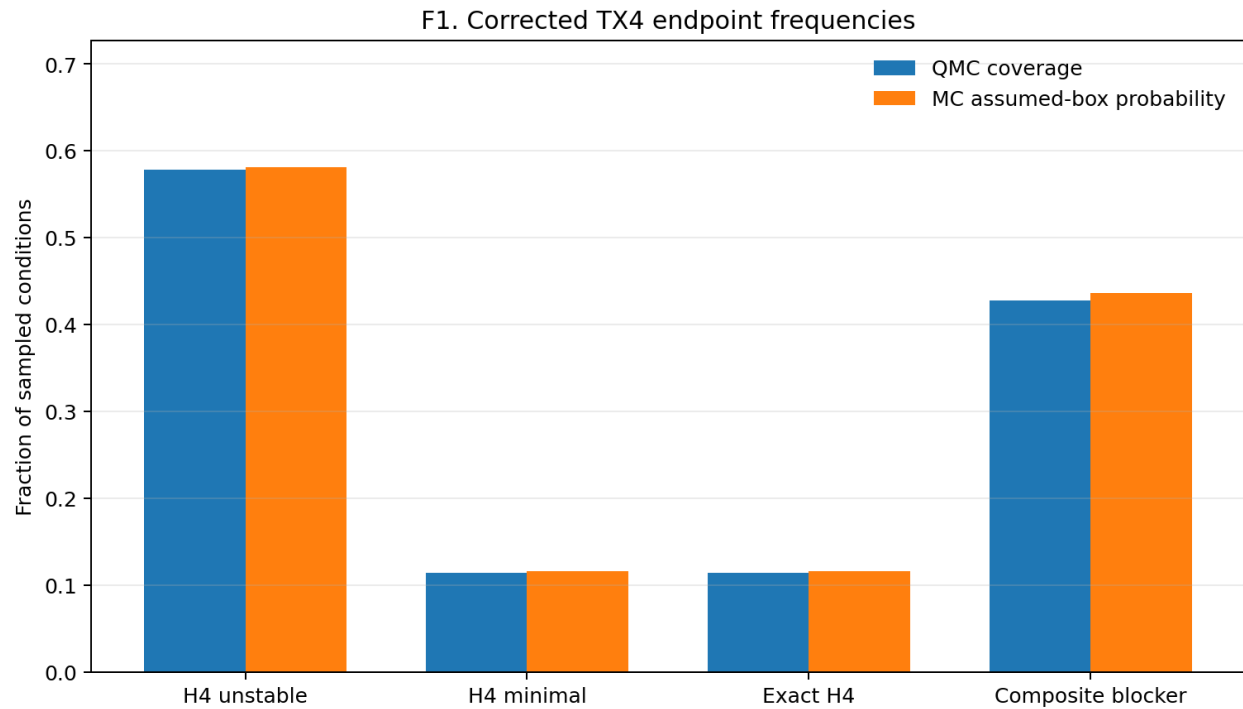
H4_PRESENT and EXACT_H4 coincide for this full-set H4 definition: once any proper unstable blocker exists, H4 cannot remain inclusion-minimal.

Continuous diagnostics

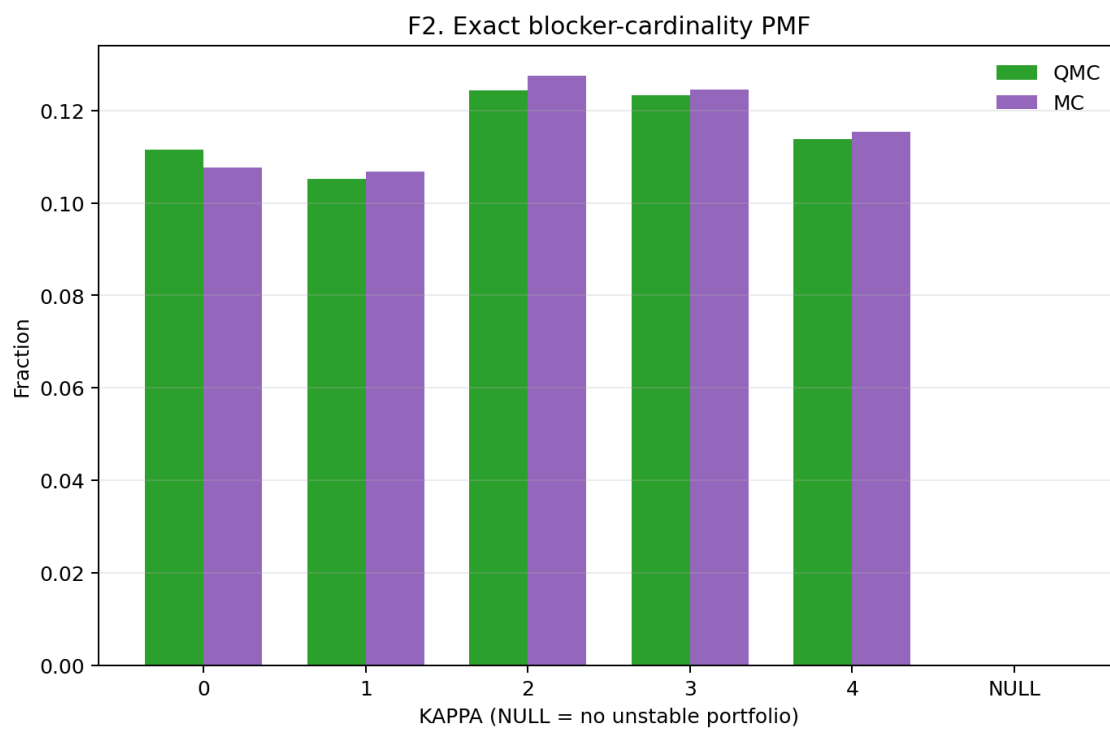
The true-minimality margin is $\min(\alpha_{EM}(H_4), -\max \text{ proper } \alpha_{EM})$. QMC median= $-0.085269 s^{-1}$, IQR= $[-0.162202, -0.028121]$, p05= -0.312240 , p95= 0.020706 . MC median= $-0.084810 s^{-1}$ with BCa 95% interval= $[-0.088820, -0.081605]$.

The exact local/collective eta stratum computed 128 of 128 requested conditions with 0 failures; all eta values were nonnegative. The definition was $\min_i \sigma_{\min}(I+Q_{\setminus i}(s_H))$.

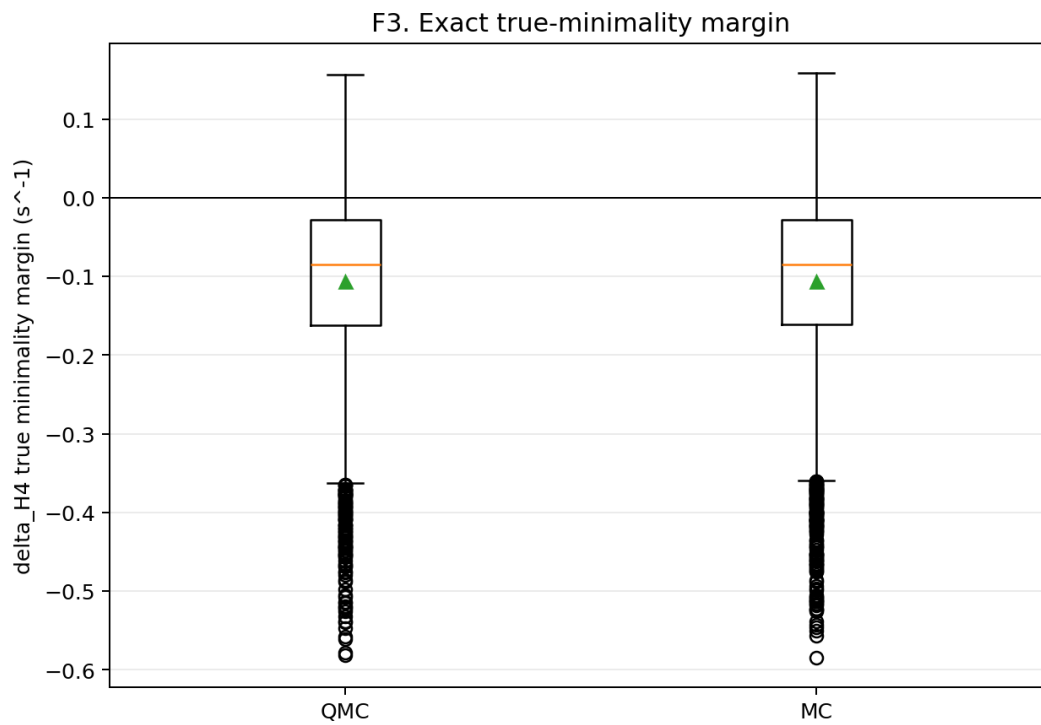
Corrected figures



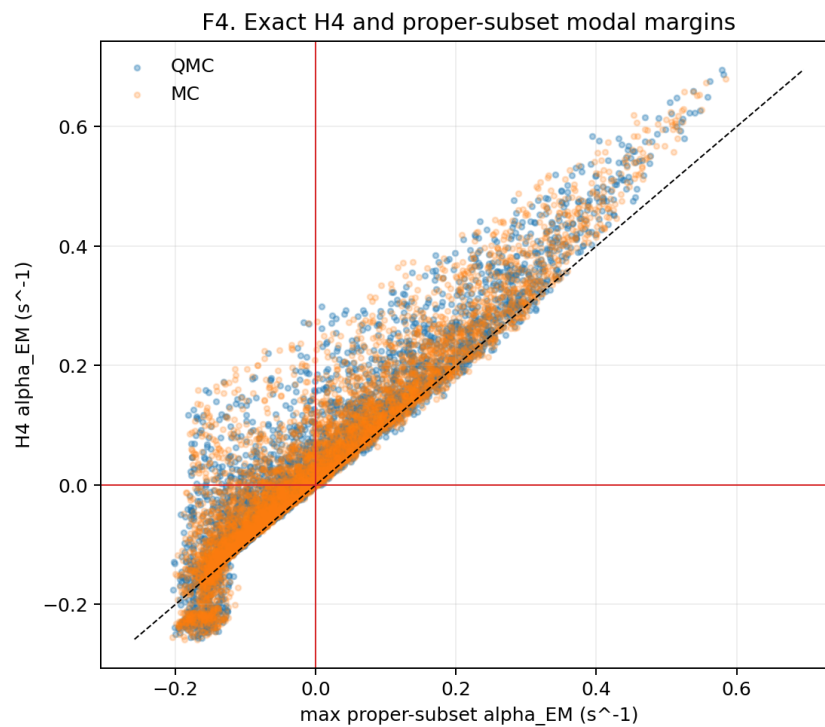
Corrected endpoint frequencies: instability, minimal H4, exact H4, and composite blockers.



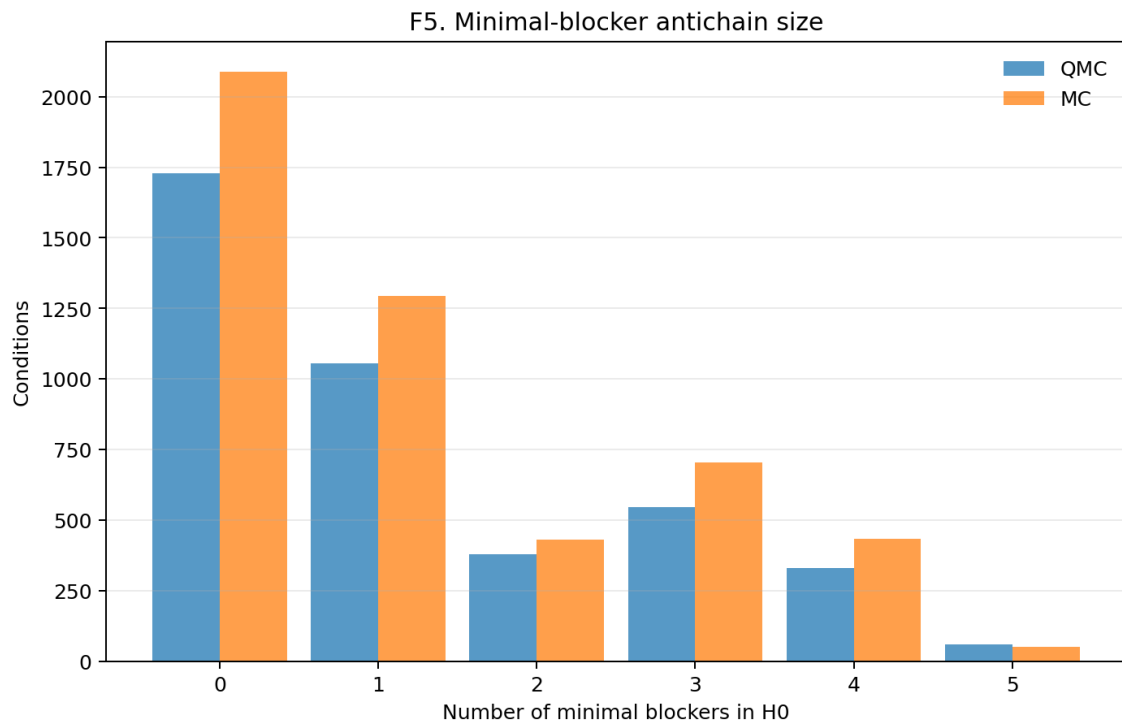
Exact KAPPA PMF, including NULL conditions.



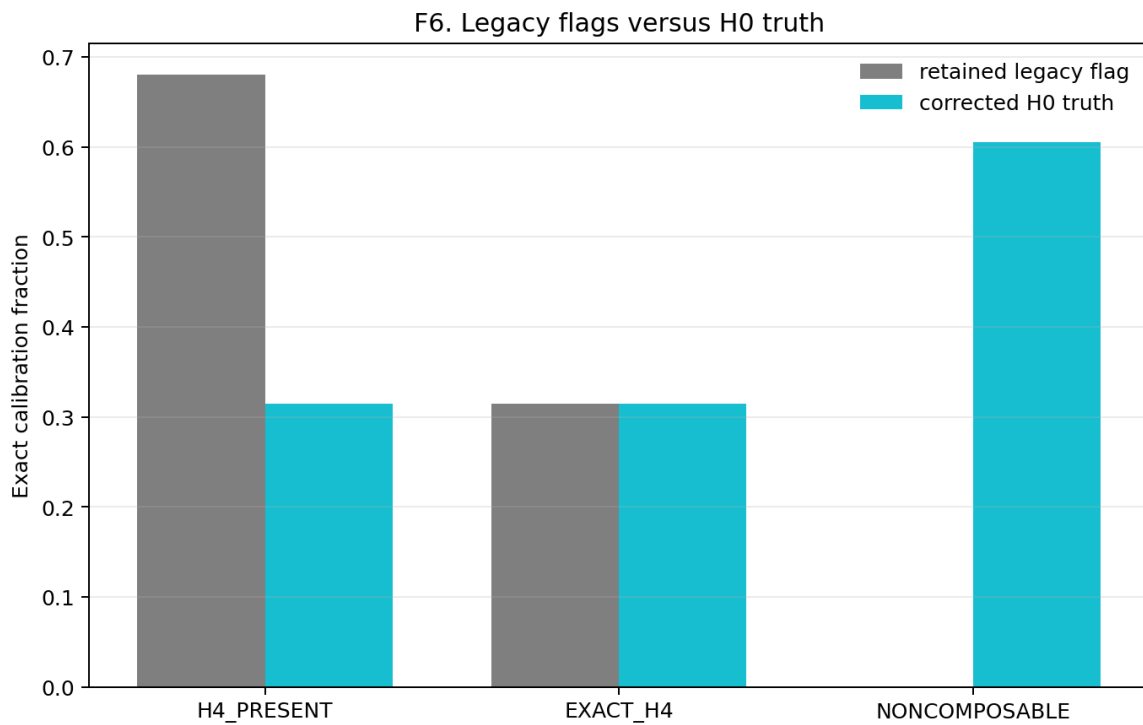
Distribution of the exact true-minimality margin.



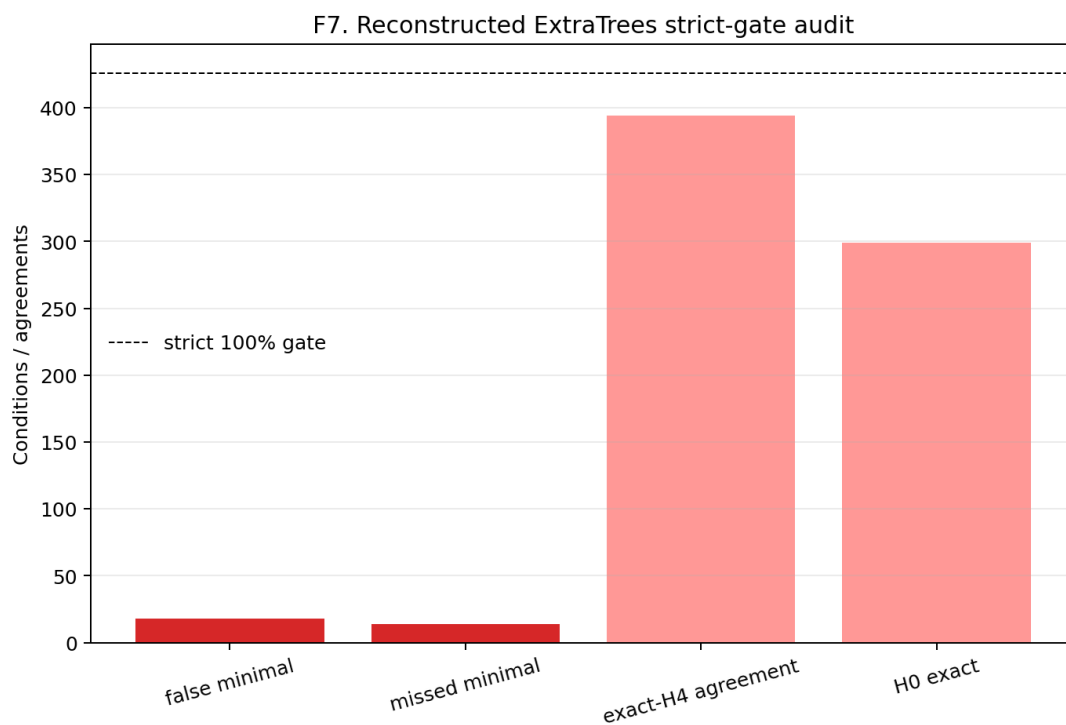
$H4 \alpha_{EM}$ against the most unstable proper subset.



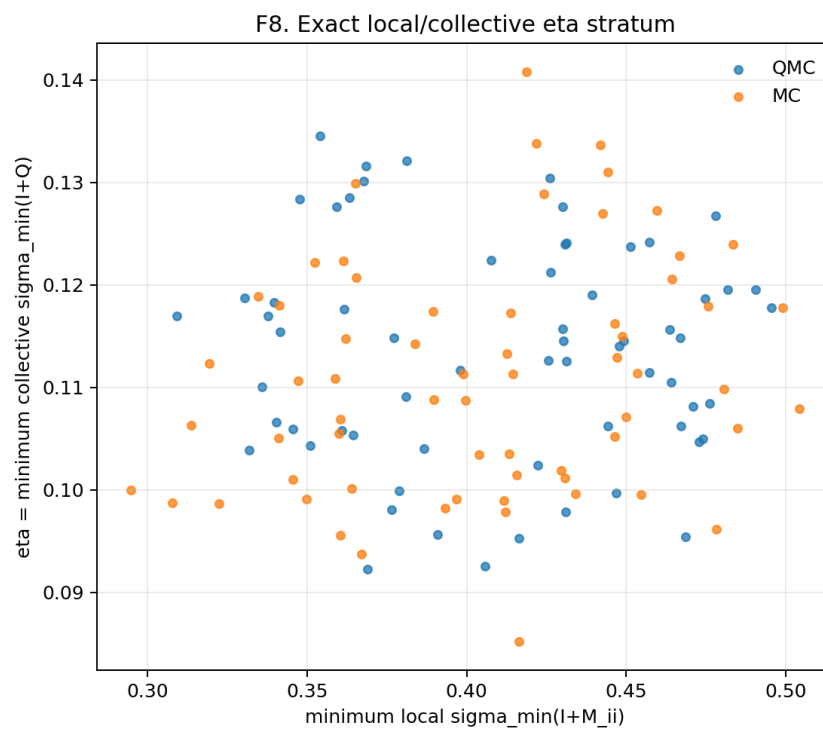
Size of the minimal-blocker antichain H0.



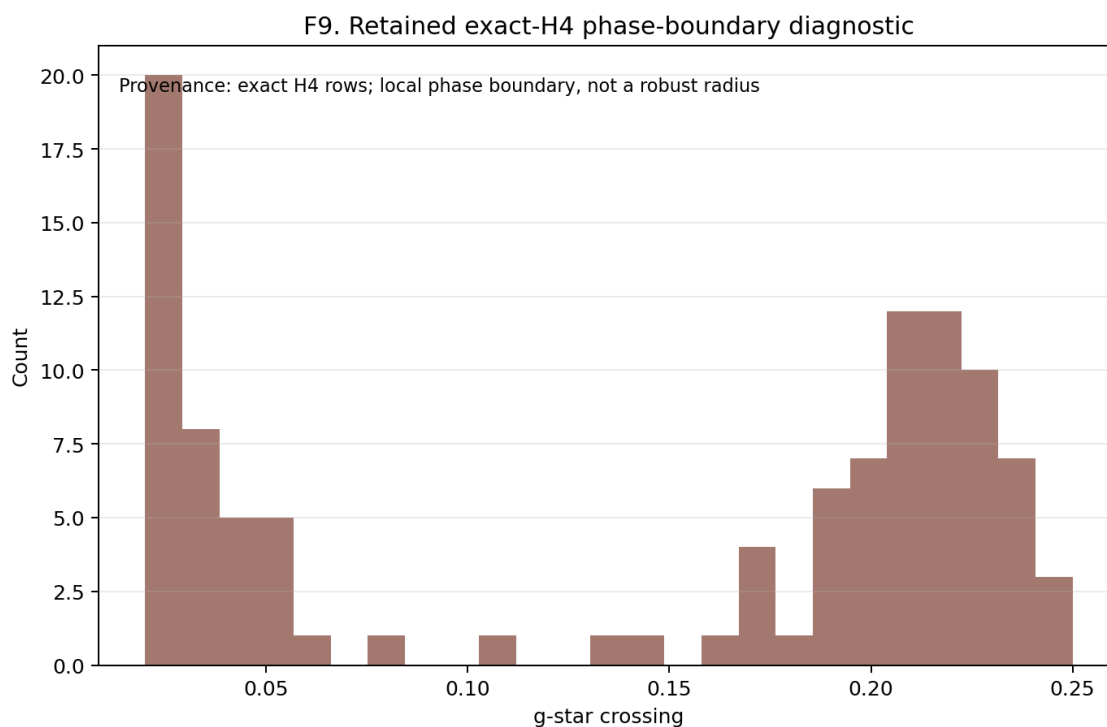
Legacy endpoint flags versus exact calibration H0 truth.



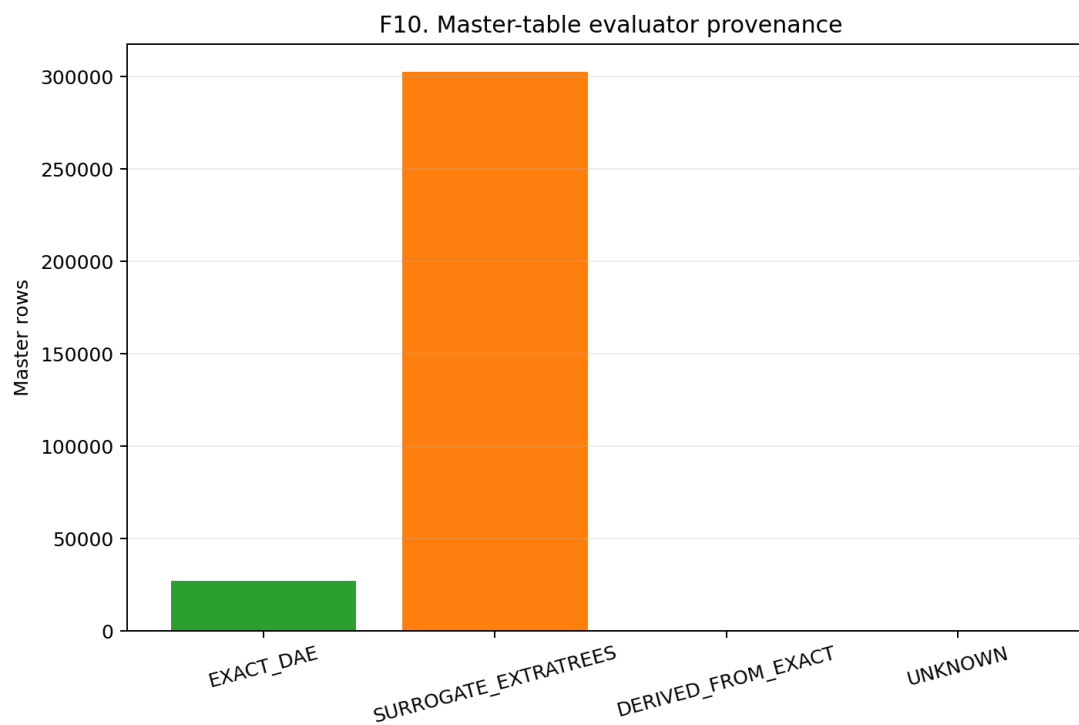
Reconstructed ExtraTrees strict-gate diagnostics.



Exact local/collective eta stratum.



Retained exact-H4 phase-boundary diagnostic with provenance.



Master-table evaluator provenance.

Audit outputs and handoff

The corrected exact Parquet files are TX4_QMC_ALL16_EXACT.parquet and TX4_MC_ALL16_EXACT.parquet. The audit retains the exact calibration truth table, blocker-list table, row-level provenance, surrogate validation/error tables, corrected eta audit, 32-check invariant table, headline JSON, and the ten corrected PNGs. The old Morris table remains labeled approximate screening and the old Sobol table remains explicitly surrogate-based; neither is promoted to exact sensitivity evidence.

The frozen exact-P4/GFL11 worktree and the original robustness branch were not modified. No push was performed. The generated paper/poster PDFs from earlier work were not used as corrected evidence.

Representative invariant status

Check	Status	Observed
QMC_row_count	PASS	65536
QMC_condition_count	PASS	4096
QMC_16_portfolios_each	PASS	4096
QMC_portfolio_catalog	PASS	16 labels
QMC_exact_source	PASS	['EXACT_DAE']
QMC_truth_condition_count	PASS	4096
QMC_implication_exact_present	PASS	4096
QMC_implication_present_unstable	PASS	4096
QMC_implication_exact_noncomposable	PASS	4096
QMC_delta_formula	PASS	allclose

Complete machine-readable invariant results are in results/TX4_ROBUSTNESS_INVARIANT_CHECKS.csv; all 32 rows are PASS.